# Joseph Casino

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## Education

**University of Illinois at Urbana-Champaign** August 2022 - May 2026
*Bachelor of Science in Computer Engineering, Minor in Business* GPA: 3.5/4.0

Coursework: Computer Systems Programming, Digital Systems Laboratory, Safe Autonomy, Sensors and Instrumentation, Robotics, Information Technology, Power Circuits and Electromechanics, Algorithms & Models of Computation

## Work Experience

**Safe-T-Sense, LLC (Euchner USA)** Schaumburg, Illinois
*Safety Services Engineer* June 2026 - Present
- Support machine safety projects across risk assessment, safety concept, engineering, verification, integration, and validation.
- Document hazards, applicable standards, safeguards, and non-compliance items to improve safety and regulatory alignment.
- Assist with engineering deliverables including I/O lists, BOMs, layouts, validation plans, and safety function documentation.

**UIUC Grainger College of Engineering** Champaign, Illinois
*Electrical and Computer Engineering (ECE) Learning Assistant* August 2024 - December 2025
- Led bi-weekly Engineering Orientation lessons on core concepts and campus resources for first-year ECE students.
- Provided 1:1 mentoring to support personal and academic development, and integration into the Grainger community.
- Collaborated with a cohort of 25 learning assistants to plan program initiatives and strengthen student retention.

**LeanMail & ScaleYOU** Barcelona, Spain
*Software Developer Automation Intern* June 2025 - August 2025
- Developed and deployed automations with Python and CRM scripting, streamlining workflows and reducing manual effort.
- Partnered with the marketing team to enhance the company website, improving performance and overall user experience.
- Contributed to end-to-end project planning, drafting technical specifications and user documentation to support delivery.

**United States Postal Service** Merrifield, Virginia
*Solutions Engineering Intern in Sortation Sytems Technology* May 2024 - July 2024
- Conducted spatial frequency response analysis on custom test targets assessing image performance of Siemen's AFCS200
- Utilized Burns Digital Imaging's ImCheck4 MATLAB Runtime software to evaluate the AFCS200 camera module quality.
- Proposed a change to the camera calibration schedule for sortation machines, backed by data analysis applicable to thousands of machines nationwide. This initiative is projected to save USPS approximately 1,600 labor hours or $132,000 annually.

## Projects

**Impact Insoles - Pressure Sensor System for Shoes** | *Embedded C, Bluetooth, PCB Design* April 2026
- Designed a wearable pressure system with 12 FSR sensors, flexible PCB insole, ESP32 MCU, 12-bit ADC, and BLE.
- Achieved 100 Hz sampling across 12 channels with <5% BLE packet loss and 28-byte data packets streamed at 2.8 kB/s.
- Validated performance through 1+ hour battery operation and cadence accuracy within ±3spm of a reference measurement.

**MTD Bus LED Tracker** | *Embedded C, STM32, Wi-Fi, PCB Design* December 2025
- Built an STM32-based PCB that visualizes real-time UIUC bus locations using the MTD API and a TLC5947 LED driver.
- Configured a WINC1500 Wi-Fi networking chip with HTTPS requests, JSON parsing, and automatic error recovery.
- Developed a scalable bus stop-to-LED mapping system with optimized full and focused scan modes to meet API rate limits.

**Real-Time Object Tracking Camera** | *Verilog, Python, Vivado, Sensors, SPI, I2C* November 2025
- Built a real-time CMV300 vision pipeline on FPGA/Python achieving 20+ FPS video and 100+ Hz IMU streaming.
- Implemented object tracking via frame differencing and centroid extraction with live visualization and control decisions.
- Integrated closed-loop motor control with sensor feedback, enabling stable 180 degree pan-tracking and smooth movement.

**Autonomous LiDAR Mapping Car** | *Python, SLAM, Robotics, LiDAR* April 2025
- Programmed an RC car equipped with a LiDAR sensor to autonomously construct a 2D map of an enclosed environment.
- Applied real-time SLAM to generate an occupancy grid, supporting dynamic mapping and accurate pose estimation.
- Integrated a breadth-first search path-planning algorithm to autonomously navigate toward the nearest unexplored region.

**RISC-V Operating System** | *RISC-V, C, OS Design* October 2024
- Architected a custom operating system with core kernel functionality, including memory management and system calls.
- Implemented virtual memory with paging, process isolation, and ELF program loading for user-space execution.
- Created virtual I/O block device driver for device communication and virtual filesystem to support unified file abstraction.

**AT-AT Walker** | *Embedded C, PWM, STM32, PCB Design* August 2024
- Engineered a Star Wars AT-AT walker replica using an STM32 microcontroller and BlueNRG-MS RF bluetooth module.
- Built a custom PCB integrating the STM32 and bluetooth chips, implementing PWM control of digital servo motors.
- Achieved robotic walking with locomotion algorithms using forward kinematics and Q matrix based motion planning.

## Skills

**Hardware**: FPGA, PCB, Ki-Cad, STM32, Xilinx Vivado/Vitis, Quartus, PWM, AXI, SPI, I2C, UART, LiDAR
**Software**: Verilog, SystemVerilog, C, C++, Assembly (RISC-V), Python, HTML/CSS, JavaScript, Git, GDB, VS Code, cPanel
**Administrative**: Microsoft Office Suites, Canva, Slack, Data Analytics, Accounting, Marketing, Organizational Management